# Whole Sentences, Exact Budgets: Retriever-Agnostic Evidence Packing for Long-Context Question Answering

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## Abstract

When a long document must be answered from under a hard token budget, most systems rank chunks or spans and concatenate the top ones until the budget runs out. We show that *how* the selected evidence is packed matters as much as *which* evidence is selected. We present SENTPACK, a training-free packer that sits behind any span ranker: it extends every selected span to the nearest sentence boundaries, merges overlapping extensions so that no text is paid for twice, admits evidence under the exact token count of the downstream model's tokenizer, keeps filling the budget with later spans after an earlier one does not fit, and emits the result in document order. We freeze every design choice on a development split and run the LongBench test set (1,150 single- and multi-document QA instances) once, with three instruction models (Qwen2.5-7B, Llama-3.1-8B, Qwen2.5-14B) and three budgets (512, 1,024, 2,048 tokens). SENTPACK outperforms dense BGE chunk packing in all nine model–budget settings, by 1.9–3.7 F1 when pooled over models ($p \leq 0.001$ at every budget), and outperforms a structured lexical pair packer by 2.5–3.1 F1. The gain is retriever-agnostic: with BGE alone as the ranker it is 2.5–2.8 F1. At 2,048 tokens, SENTPACK is statistically indistinguishable from reading the full context on all three models, and it exceeds LLMLingua-2 at the same budget by 15.3 F1.

## 1 Introduction

Large language models (LLMs) are routinely asked to answer questions about long documents and document collections. Even when the context window is long enough, there are reasons to send the model less text: prefill cost and latency grow with prompt length, many deployed models have short windows, and long inputs dilute the few passages that matter (Liu et al., 2024a). In practice a long context is therefore often reduced to a *hard* budget of a few hundred to a few thousand tokens, measured in the tokenizer of the model that will read it.

Two families of methods do this. *Prompt compressors* (Li et al., 2023; Jiang et al., 2023, 2024; Pan et al., 2024) delete tokens or phrases judged uninformative. *Retrieval-style packers* split the context into chunks or spans, rank them with BM25 or a dense encoder (Robertson and Zaragoza, 2009; Karpukhin et al., 2020; Xiao et al., 2023b), and concatenate the top-ranked units, each padded with a fixed window of neighbouring tokens, until the budget is exhausted. Research effort has concentrated on the ranking step. The packing step is usually treated as bookkeeping, and it has three habits that are costly under a tight budget: fixed windows cut sentences, and frequently the answer phrase, at arbitrary positions; adjacent or overlapping units repeat the same text and pay for it twice; and greedy admission stops filling, or leaves holes, as soon as one long unit does not fit.

We argue that under an exact budget the packer deserves the same care as the ranker, and we present SENTPACK, a packer that can be placed behind any span ranker (Figure 1). SENTPACK (i) extends each selected span to the nearest sentence boundaries within a small radius, (ii) merges extensions that touch or overlap into a single passage, (iii) admits spans in rank order under the exact downstream token count and *backfills*, i.e., keeps admitting later spans after one does not fit, and (iv) emits the merged passages verbatim in document order. None of these operations requires training, and all of them run on the CPU before the model is called.

We evaluate SENTPACK under a protocol designed to leave little room for doubt. Every design choice, including the ranker, was fixed on a development split built from LongBench-E with all LongBench test questions removed; the LongBench test set was then run once. All methods share the candidate spans, the prompt wrapper, the

model, greedy decoding, the scorer, and a budget counted with the downstream tokenizer. We test three instruction models and three budgets and report paired bootstrap tests. Our findings are:

- **Packing beats ranking-only improvements.** SENTPACK outperforms dense BGE chunk packing in all nine model–budget settings, by +3.68, +1.92 and +2.26 F1 at 512, 1,024 and 2,048 tokens (pooled over models, all $p \leq 0.001$), and outperforms a structured lexical pair packer by 2.5–3.1 F1 (Tables 1 and 2).
- **The gain is retriever-agnostic.** Holding the BGE ranking fixed and changing only the packer improves F1 by 2.5–2.8 at every budget; with a purely lexical ranker SENTPACK still outperforms the pair packer that uses the same lexical signals (Section 5.2).
- **A 2,048-token SENTPACK context is as good as the full context** on all three models (paired differences $+0.56$, $-0.35$ and $-0.10$ F1, none significant), while sending about one sixth of the prompt.
- **Evidence-level packing dominates token deletion.** At the same exact budget, SENTPACK exceeds LLMLingua-2 by 15.3 F1.
- **The packer is robust.** Removing any one of its components still leaves a packer that beats BGE chunk packing; merging overlapping evidence is the component whose removal costs the most (Section 5.5).

## 2 Related Work

**Prompt and context compression.** Selective Context removes lexical units with low self-information (Li et al., 2023). LLMLingua and LongLLMLingua estimate token importance with a small LM, the latter with question-aware coarse-to-fine compression (Jiang et al., 2023, 2024); LLMLingua-2 distils keep/drop labels into a bidirectional encoder (Pan et al., 2024). CPC scores sentences against the question with a trained encoder (Liskavets et al., 2025), and RECOMP and CompAct train extractive or abstractive compressors (Xu et al., 2024; Yoon et al., 2024). SENTPACK is training-free, never edits text inside a kept passage, and is complementary to learned scorers: any of them can supply its ranking.

**Retrieval-style context construction.** Retrieval-augmented generation ranks passages with sparse (Robertson and Zaragoza, 2009) or dense (Karpukhin et al., 2020; Xiao et al., 2023b) retrievers and concatenates the top results (Lewis et al., 2020; Izacard and Grave, 2021); diversity-aware selection (Carbonell and Goldstein, 1998) and multi-hop retrieval (Xiong et al., 2021; Trivedi et al., 2023) refine *which* units are selected. We study the complementary question of how selected units are turned into a context under an exact budget, and show that this step alone moves F1 by 2–4 points.

**Long-context evaluation and KV-cache compression.** LongBench (Bai et al., 2023) evaluates QA over long inputs, and models under-use evidence in the middle of long contexts (Liu et al., 2024a). KV-cache methods evict, merge or quantise cached states during generation (Xiao et al., 2023a; Zhang et al., 2023; Li et al., 2024; Liu et al., 2024b). SENTPACK instead shortens the input before prefill, so the full context is never encoded.

## 3 Method

### 3.1 Problem Setting

Given a question $q$ and a long context $C$, we construct $\hat{C}$ with $|\hat{C}|_\tau \leq B$, where $|\cdot|_\tau$ counts tokens with the tokenizer $\tau$ of the downstream model. A fixed prompt wrapper keeps the task instruction and question unchanged; only the context slot is replaced by $\hat{C}$. Because the budget is defined in downstream tokens, every method is held to exactly the same constraint for each model.

### 3.2 Candidate Spans

The context is split into contiguous pages of 256 regex tokens (words or single punctuation marks), and each page into candidate spans of 64 regex tokens with stride 32; spans never cross pages. Regex tokens only define candidate units; all budget accounting uses $\tau$. All span-based methods in our comparison use the same candidate spans.

### 3.3 Rankers

SENTPACK accepts any function that scores spans against $q$. We use three, none of which sees gold answers. The *lexical* ranker fuses $z$-normalised BM25 ($k_1$=1.2, $b$=0.75) and TF-IDF scores computed over the spans of the instance. The *dense* ranker is the cosine similarity between the question (with BGE's query instruction) and the span under

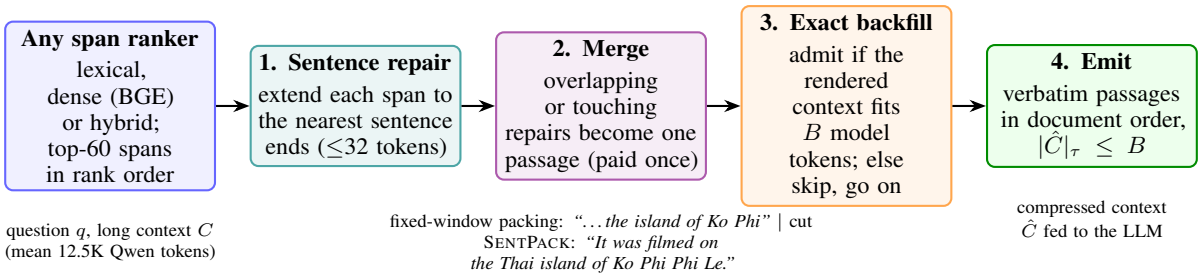



Figure 1: SENTPACK packs the spans returned by any ranker. Each span is extended to sentence boundaries, overlapping extensions are merged, spans are admitted in rank order under the exact downstream token budget with backfilling, and the result is emitted verbatim in document order. SENTPACK runs before prefill and does not modify the model.

`bge-small-en-v1.5` (Xiao et al., 2023b). The *hybrid* ranker adds the $z$-normalised lexical and dense scores. The top 60 spans form the priority list $P$, ordered by score with ties broken by position. Our main configuration uses the hybrid ranker, chosen on the development split; Section 5.2 shows the gains do not depend on this choice.

### 3.4 The Packer

**Sentence repair.** Fixed-size spans start and end at arbitrary positions. Rather than padding each span by a fixed window, SENTPACK extends its left end backwards to the first token after a sentence terminator (`.`, `?`, `!`) and its right end forwards to the next terminator, each by at most $r = 32$ regex tokens and within the page. A repaired span is thus never longer than a fixed $\pm r$ window, and usually shorter, but it starts and ends on sentence boundaries whenever one is within reach.

**Merging.** Spans that are close in the document produce repaired ranges that overlap or touch. SENTPACK sorts the repaired ranges of the current selection by page and position and merges every pair that overlaps into one passage. Overlapping evidence is therefore rendered, and paid for, once.

**Exact admission with backfill.** Spans are considered in priority order. A span is admitted if the context rendered from the current selection plus the span (repaired, merged and joined after the prefix "Question: $q$ / Selected evidence:") has at most $B$ tokens under $\tau$. Because merging can make a new span nearly free, the cost of a span depends on what is already selected, so it is measured on the rendered context rather than estimated. If a span does not fit, SENTPACK skips it and continues with the next one, so short spans further down the list can still use the remaining budget.

**Document-order emission.** The merged passages are emitted verbatim in document order, separated by blank lines. If re-joining changes the token count, the context is clipped to $B$, so $|\hat{C}|_\tau \leq B$ always holds. Algorithm 1 (Appendix A) gives the procedure.

**Development protocol.** All choices—the ranker, sentence rather than fixed repair, $r$, the list length, merging, backfilling and emission order—were made on a development split of 584 instances: the HotpotQA and 2WikiMultihopQA portions of LongBench-E (Bai et al., 2023) after removing every question that also appears in the LongBench test set. The development experiments used a 4-bit Qwen2.5-7B model on a separate GPU. The configuration was then frozen and the test set was run once with every model and budget reported here; no test result was used to change the method.

## 4 Experimental Setup

**Data.** All English single- and multi-document QA tasks of LongBench (Bai et al., 2023): Qasper (Dasigi et al., 2021) (200), MultiFieldQA-en (150), HotpotQA (Yang et al., 2018) (200), 2WikiMultihopQA (Ho et al., 2020) (200), MuSiQue (Trivedi et al., 2022) (200) and NarrativeQA (Kočiský et al., 2018) (200), 1,150 instances. We report LongBench's F1 and the macro average over the six tasks; "MH" averages the three multi-hop tasks.

**Models and decoding.** Qwen2.5-7B-Instruct, Qwen2.5-14B-Instruct (Qwen Team, 2024) and Llama-3.1-8B-Instruct (Grattafiori et al., 2024), in bfloat16 with vLLM, LongBench's task prompts inside each model's chat template, greedy decoding and LongBench's per-task output lengths. The 14B model reads the packed contexts built with the Qwen tokenizer, which it shares with the 7B model.

**Budgets.** $B \in \{512, 1{,}024, 2{,}048\}$ downstream tokens for the context slot. The mean full prompt has 12.5K Qwen tokens, so these budgets keep roughly 4%, 8% and 16% of the context.

**Baselines.** All baselines fill the same slot under the same exact budget and tokenizer. *BGE chunks*: the 64 spans most similar to the question under BGE, each padded by $\pm 32$ tokens, admitted in similarity order while they fit and emitted in that order; this is the standard dense packing recipe. *Pair packing*: a structured lexical packer that selects query–support span pairs with the lexical ranker, a lexical bridge score and a redundancy penalty, repairs each span with a fixed $\pm 32$ window, admits pairs until the first one that does not fit and emits them in document order. It represents structured, multi-hop-aware selection with conventional packing. *LLMLingua-2* (Pan et al., 2024) (`xlm-roberta-large-meetingbank`), asked to compress the context to the budget; its output is placed after the same prefix and truncated to $B$ downstream tokens if needed. *Full context*: the whole context, middle-truncated to 30K tokens as in LongBench (a 32K window for all models), as a reference. Appendix E compares the pair packer with truncation, Selective Context, LongLLMLingua and BM25/TF-IDF chunk packing.

**Statistics.** Comparisons are paired on instances. We use a task-stratified paired bootstrap (Efron and Tibshirani, 1993) with 10,000 resamples and seed 20260701; the statistic is the macro average of per-task mean differences. Pooled results resample within the eighteen model–task strata. We report 95% percentile intervals and two-sided $p$-values.

# 5 Results

## 5.1 SENTPACK versus standard packing

Table 1 gives macro F1 for every model and budget, and Table 2 the pooled paired tests. SENTPACK outperforms BGE chunk packing in all nine model–budget settings. Pooled over models, the gain is +3.68 F1 at 512 tokens, +1.92 at 1,024 and +2.26 at 2,048, and every interval excludes zero. Per model, eight of the nine differences are significant at $p < 0.05$ (the ninth, Qwen2.5-7B at 1,024 tokens, is +1.79 with $p = 0.06$; Appendix D). The gain is largest at the smallest budget, where every wasted token is most costly.

SENTPACK also outperforms the structured pair packer by 2.5–3.1 F1 at every budget ($p < 0.001$), in all nine settings. The pair packer was designed for multi-hop questions and selects evidence in pairs, yet a packer that ranks spans one at a time but packs them carefully is better on the multi-hop tasks as well: at 1,024 tokens SENTPACK reaches 46.3, 43.9 and 50.3 MH F1 on the three models, against 42.4, 39.6 and 47.0 for pair packing and 42.7, 41.3 and 48.0 for BGE chunks (Appendix C).

## 5.2 The packer, not the ranker, carries the gain

The most direct test of the packer is to keep the ranking fixed. BGE chunk packing and SENTPACK with the BGE ranker use the same BGE similarities over the same spans and differ only in how the top spans are turned into a context. Changing the packer improves F1 by +2.81, +2.48 and +2.57 at the three budgets (Table 2), and per model it is significant in all nine settings (Appendix D).

Conversely, the choice of ranker inside SENTPACK matters little. The hybrid ranker selected on the development split and the BGE ranker are within 0.6 F1 of each other on average and their difference is not significant ($-0.56$, $[-1.49, 0.40]$ at 1,024 tokens); each is best in some settings. With a purely lexical ranker, SENTPACK outperforms the pair packer, which uses the same lexical signals, by 1.54 F1 [0.68, 2.43] ($p < 0.001$) at 1,024 tokens. In short, the improvement comes from the packing operations and transfers across lexical, dense and hybrid rankers.

## 5.3 How close to the full context?

At 2,048 tokens, SENTPACK is statistically indistinguishable from reading the full context on all three models: the paired differences are $+0.56$ $[-1.46, 2.56]$ for Qwen2.5-7B, $-0.35$ $[-2.39, 1.65]$ for Llama-3.1-8B and $-0.10$ $[-2.04, 1.82]$ for Qwen2.5-14B. The packed context is about one sixth of the mean prompt (2,048 vs. 12.5K tokens). On the multi-hop tasks it exceeds the full context on all three models (49.2 vs. 44.4, 45.8 vs. 43.7 and 56.0 vs. 52.6 MH F1), consistent with the observation that long inputs dilute the evidence that matters (Liu et al., 2024a). At 1,024 tokens, SENTPACK recovers 93–95% of full-context F1.

## 5.4 Evidence-level vs. token-level compression

Under the same exact 1,024-token budget, LLMLingua-2 reaches 24.6–28.4 F1, and SENTPACK exceeds it by 15.27 F1 [13.90, 16.64] pooled

| Method | Qwen2.5-7B | | | Llama-3.1-8B | | | Qwen2.5-14B | | | Avg. |
|---|---|---|---|---|---|---|---|---|---|---|
| | 512 | 1,024 | 2,048 | 512 | 1,024 | 2,048 | 512 | 1,024 | 2,048 | |
| Full context (reference) | | 43.04 | | | 43.13 | | | 47.59 | | |
| LLMLingua-2 | – | 24.60 | – | – | 28.43 | – | – | 27.10 | – | – |
| Pair packing | 33.88 | 39.11 | 40.68 | 34.05 | 37.79 | 39.60 | 36.73 | 41.69 | 44.23 | 38.64 |
| BGE chunks | 33.05 | 38.85 | 40.98 | 33.27 | 38.99 | 40.85 | 36.27 | 42.34 | 45.29 | 38.88 |
| SENTPACK (BGE ranker) | 35.64 | **40.76** | 43.14 | 36.16 | **41.67** | **43.58** | 39.21 | **45.18** | **48.10** | **41.49** |
| SENTPACK (hybrid ranker, main) | **36.74** | 40.64 | **43.60** | **36.44** | 40.98 | 42.78 | **40.44** | 44.32 | 47.50 | **41.49** |

Table 1: Macro F1 on the 1,150 LongBench QA instances under exact budgets of 512, 1,024 and 2,048 downstream tokens. The full-context reference reads up to 30K tokens. Bold marks the best compressed result in each column. Per-task scores are in Appendix C.

| SENTPACK minus | 512 | 1,024 | 2,048 |
|---|---|---|---|
| BGE chunks | **+3.68** [2.53, 4.82] | **+1.92** [0.87, 2.97] | **+2.26** [1.24, 3.29] |
| Pair packing | **+2.99** [2.04, 3.94] | **+2.45** [1.52, 3.39] | **+3.12** [2.18, 4.08] |
| *Same BGE ranking, different packer:* | | | |
| SENTPACK-BGE − BGE chunks | **+2.81** [1.88, 3.76] | **+2.48** [1.50, 3.45] | **+2.57** [1.58, 3.57] |

Table 2: Paired F1 differences pooled over the three models (95% bootstrap intervals). All intervals exclude zero ($p \leq 0.001$). Per-model tests are in Appendix D.

| Variant (1,024 tokens) | vs. BGE chunks | Full − variant |
|---|---|---|
| SENTPACK (all components) | +1.92* | – |
| without merging | +1.14* | +0.78* [0.03, 1.52] |
| without backfill | +1.63* | +0.29 [−0.13, 0.71] |
| fixed ±32 repair | +1.92* | +0.00 [−0.75, 0.73] |
| rank-order emission | +2.14* | −0.22 [−0.97, 0.53] |

Table 3: Component analysis, pooled over the three models (F1, paired bootstrap; *: 95% interval excludes zero). Each variant switches off one operation and keeps the hybrid ranker and the exact budget.

over models. At roughly 12× compression a token-level compressor must remove most of every sentence, including the names, numbers and function words that extractive QA needs to copy, whereas SENTPACK spends the budget on a few complete passages. Appendix E shows the same ordering for Selective Context and LongLLMLingua.

### 5.5 What each packing operation contributes

Table 3 switches off one packing operation at a time. Two observations stand out. First, the packer is robust: every single-component variant still outperforms BGE chunk packing significantly, so no single operation is a point of failure. Second, merging overlapping evidence is the operation whose removal costs the most (+0.78 F1, $p = 0.04$). Without merging, two selected spans from the same passage are rendered as two overlapping copies, and the duplicated text displaces other evidence; with merging, a span adjacent to already selected evidence costs only its new tokens, which also makes exact, rendered-context admission necessary. The remaining operations are interchangeable in the sense that one can compensate for another: sentence repair and fixed-window repair perform the same once merging and backfilling are in place, and so do document-order and rank-order emission. On the development split, sentence repair improved F1 by 2.5 over fixed repair with a lexical ranker (Appendix B); we keep it because it never lengthens a span beyond the fixed window and yields readable passages.

### 5.6 Cost

SENTPACK runs on a single CPU core in 432 ms per instance on average (median 415 ms, 95th percentile 698 ms, 1,024 tokens, 300 random test instances), compared with 614 ms for the pair packer and 280 ms for BGE chunk packing; the BGE encoder pass is shared by both dense methods. Every packed context respects the exact budget (maximum 1,024 tokens), and SENTPACK uses the budget more fully than chunk packing (1,014 vs. 987 tokens on average) because backfilling closes the gaps left by spans that do not fit.

## 6 Discussion

**Why does packing matter so much?** Under an exact budget, every token spent on a broken sentence, a duplicated window or an unfilled gap is a token not spent on evidence. Standard packing pays all three costs. Sentence repair turns spans into self-contained statements, merging stops the same text from being bought twice, backfilling uses the remainder of the budget, and document order keeps neighbouring passages in their natural sequence. Each effect is small, and the operations

partly substitute for one another, but together they are worth 2–4 F1, as much as replacing a lexical ranker with a dense one.

**Implications for compression research.** Most work on context compression compares rankers or learned scorers with a fixed, simple packer. Our results suggest that such comparisons should hold the packer fixed *and* strong, because a better packer gives every ranker a similar boost. SENTPACK needs only the downstream tokenizer and adds no model calls, so it can be adopted as the default packer behind any retriever or learned scorer.

**Deployment.** SENTPACK is training-free, runs before the model is called and outputs verbatim source text in document order, so the compressed context can be shown to users and traced back to its sources.

## 7 Conclusion

We studied long-context question answering under a hard budget measured in the downstream model's own tokens and showed that the step that turns ranked spans into a context is a major, understudied lever. SENTPACK—sentence repair, merging of overlapping evidence, exact admission with backfill and document-order emission—improves over dense chunk packing in all nine model–budget settings we test, by 1.9–3.7 F1 pooled over models, transfers across lexical, dense and hybrid rankers, and at 2,048 tokens matches the full context on three LLMs. We hope this encourages treating packing as a first-class component of context compression.

## Limitations

SENTPACK shortens the input before prefill and is not a KV-cache method. Our experiments cover English extractive QA on LongBench with three instruction models of 7–14B parameters and budgets of 512–2,048 tokens; other languages, generation tasks such as summarisation, and much larger models may behave differently. The development split covers two multi-hop tasks, and the method was frozen before testing; a broader development set might lead to a different choice of ranker, although our results suggest the ranker matters less than the packer. Our component analysis shows that the packing operations partly substitute for one another, so their individual contributions are smaller than their joint effect. The dense and hybrid rankers require a small encoder at inference time.

## Ethics Statement

Context compression can remove information a user would consider important, which can lead to incomplete or incorrect answers; in high-stakes settings, compressed contexts should be shown to the user or checked against the source. SENTPACK keeps all admitted text verbatim, which makes such checks straightforward. All datasets are public research benchmarks used under their licences.

## References


Yushi Bai, Xin Lv, Jiajie Zhang, Hongchang Lyu, Jiankai Tang, Zhidian Huang, Zhengxiao Du, Xiao Liu, Aohan Zeng, Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. 2023. Longbench: A bilingual, multitask benchmark for long context understanding. *arXiv preprint arXiv:2308.14508*.

Jaime Carbonell and Jade Goldstein. 1998. The use of MMR, diversity-based reranking for reordering documents and producing summaries. In *Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 335–336.

Pradeep Dasigi, Kyle Lo, Iz Beltagy, Arman Cohan, Noah A. Smith, and Matt Gardner. 2021. A dataset of information-seeking questions and answers anchored in research papers. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4599–4610.

Bradley Efron and Robert J. Tibshirani. 1993. *An Introduction to the Bootstrap*. Chapman and Hall/CRC.

Aaron Grattafiori et al. 2024. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. 2020. Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 6609–6625.

Gautier Izacard and Edouard Grave. 2021. Leveraging passage retrieval with generative models for open domain question answering. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics*, pages 874–880.

Huiqiang Jiang, Qianhui Wu, Chin-Yew Luo, Dongsheng Li, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. 2023. Llmlingua: Compressing prompts for accelerated inference of large language models. *arXiv preprint arXiv:2310.05736*.

Huiqiang Jiang, Qianhui Wu, Chin-Yew Luo, Dongsheng Li, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. 2024. Longllmlingua: Accelerating and enhancing llms in long context scenarios via prompt compression. *arXiv preprint arXiv:2310.06839*.

Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 6769–6781.

Tomáš Kočiský, Jonathan Schwarz, Phil Blunsom, Chris Dyer, Karl Moritz Hermann, Gábor Melis, and Edward Grefenstette. 2018. The NarrativeQA reading comprehension challenge. *Transactions of the Association for Computational Linguistics*, 6:317–328.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen-tau Yih, Tim Rocktaschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474.

Yucheng Li, Bo Dong, Chenghua Lin, and Frank Guerin. 2023. Selective context: Efficient context selection for large language models. *arXiv preprint arXiv:2304.12102*.

Yucheng Li, Ying Huang, Bowen Yang, Bharat Venkitesh, Andrea Locatelli, Hanchen Ye, Tianle Cai, Patrick Lewis, and Deming Chen. 2024. Snapkv: Llm knows what you are looking for before generation. *arXiv preprint arXiv:2404.14469*.

Barys Liskavets, Maxim Ushakov, Shuvendu Roy, Mark Klibanov, Ali Etemad, and Shane K. Luke. 2025. Prompt compression with context-aware sentence encoding for fast and improved LLM inference. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 24595–24604.

Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024a. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173.

Zirui Liu, Jiayi Yuan, Hongye Jin, Shaochen Zhong, Zhaozhuo Xu, Vladimir Braverman, Beidi Chen, and Xia Hu. 2024b. Kivi: A tuning-free asymmetric 2bit quantization for kv cache. *arXiv preprint arXiv:2402.02750*.

Zhuoshi Pan, Qianhui Wu, Huiqiang Jiang, Menglin Xia, Xufang Luo, Jue Zhang, Qingwei Lin, Victor Rühle, Yuqing Yang, Chin-Yew Lin, H. Vicky Zhao, Lili Qiu, and Dongmei Zhang. 2024. LLMLingua-2: Data distillation for efficient and faithful task-agnostic prompt compression. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 963–981.

Qwen Team. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Stephen Robertson and Hugo Zaragoza. 2009. The probabilistic relevance framework: Bm25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2022. MuSiQue: Multi-hop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2023. Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, pages 10014–10037.

Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. 2023a. Efficient streaming language models with attention sinks. *arXiv preprint arXiv:2309.17453*.

Shitao Xiao, Zheng Liu, Peitian Zhang, and Niklas Muennighoff. 2023b. C-pack: Packaged resources to advance general Chinese embedding. *arXiv preprint arXiv:2309.07597*.

Wenhan Xiong, Xiang Lorraine Li, Srinivasan Iyer, Jingfei Du, Patrick Lewis, William Yang Wang, Yashar Mehdad, Wen-tau Yih, Sebastian Riedel, Douwe Kiela, and Mohit Bansal. 2021. Answering complex open-domain questions with multi-hop dense retrieval. In *Proceedings of the International Conference on Learning Representations*.

Fangyuan Xu, Weijia Shi, and Eunsol Choi. 2024. Recomp: Improving retrieval-augmented lms with compression and selective augmentation. In *Proceedings of the International Conference on Learning Representations*.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. 2018. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2369–2380.

Chanwoong Yoon, Taewhoo Lee, Hyeon Hwang, Minbyul Jeong, and Jaewoo Kang. 2024. CompAct: Compressing retrieved documents actively for question answering. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 21424–21439.

Zhenyu Zhang, Ying Sheng, Tianyi Zhou, Tianlong Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Re, Clark Barrett, Zhangyang Wang, and Beidi Chen. 2023. H2o: Heavy-hitter oracle for efficient generative inference of large language models. *arXiv preprint arXiv:2306.14048*.

# A Implementation Details

**Algorithm 1** SENTPACK

**Require:** question $q$, context $C$, budget $B$, tokenizer $\tau$, ranker $f$
split $C$ into 256-token pages and 64-token spans (stride 32)
$P \leftarrow$ top 60 spans by $f(q, s)$
$\mathcal{S} \leftarrow \emptyset$
**for** $s \in P$ in order **do**
  $R \leftarrow$ sentence-repaired ranges of $\mathcal{S} \cup \{s\}$ ($\leq$32 tokens per side, within page)
  merge overlapping ranges of $R$; render in document order after the prefix
  **if** $|\text{render}|_\tau \leq B$ **then** $\mathcal{S} \leftarrow \mathcal{S} \cup \{s\}$
  **end if** ▷ otherwise skip $s$ and continue (backfill)
**end for**
**return** render of $\mathcal{S}$, clipped to $B$ tokens if needed

| Item | Value |
|---|---|
| Pages / spans | 256 regex tokens / 64, stride 32 |
| Lexical ranker | $0.5\,z(\text{BM25}) + 0.5\,z(\text{TFIDF})$; $k_1$=1.2, $b$=0.75; sublinear TF, smoothed IDF |
| Dense ranker | `bge-small-en-v1.5`, CLS pooling, $\ell_2$-normalised, query instruction |
| Hybrid ranker | $z(\text{lexical}) + z(\text{dense})$ |
| Priority list | top 60 spans |
| Sentence terminators | . ? ! |
| Repair radius | 32 regex tokens per side, within page |
| Admission | rendered context under $\tau$, backfill |
| Emission | document order, blank-line separated |

Table 4: Implementation constants (all frozen before testing).

Inference used vLLM 0.7.3 with bfloat16 weights on NVIDIA A100-40GB GPUs (tensor parallelism of two for the 14B full-context run), with a 4,096-token model window for compressed contexts and 32,768 tokens for the full-context reference. Code, packed contexts and per-instance predictions for every number in this paper will be released.

# B Development Results

Table 5 reports macro F1 on the development split (HotpotQA-E and 2WikiMultihopQA-E without test questions, 584 instances, 1,024 tokens, 4-bit Qwen2.5-7B-Instruct). These results determined the frozen configuration; the hybrid ranker with sentence repair was the best development configuration, +5.26 F1 [2.34, 8.23] above BGE chunk packing.

| Ranker (all with merge + backfill) | Fixed $\pm$32 | Sentence |
|---|---|---|
| Lexical | 45.37 | 47.90 |
| BGE | 48.88 | 50.12 |
| Hybrid | 48.17 | **51.19** |
| BGE chunks | 45.93 | |
| Pair packing | 45.03 | |

Table 5: Development macro F1 (1,024 tokens).

# C Per-Task Results

Table 6 lists per-task F1 at 1,024 tokens. Multi-hop (MH) F1 of SENTPACK (hybrid) at 512 / 2,048 tokens is 39.47 / 49.20 (Qwen2.5-7B), 37.02 / 45.78 (Llama-3.1-8B) and 44.18 / 56.01 (Qwen2.5-14B), against 34.84 / 46.37, 33.52 / 44.14 and 40.02 / 52.57 for BGE chunks.

# D Per-Model Paired Tests

# E Additional Baselines

In an earlier run with the same candidate spans, prompt wrapper, scorer and exact 1,024-token budget, but with Qwen2.5-7B-Instruct and Meta-Llama-3-8B-Instruct in an earlier software environment, the pair packer of Table 1 outperformed head and middle truncation by 13.1 and 13.6 F1, Selective Context by 18.2, LLMLingua-2 by 11.3, LongLLMLingua by 6.6, and BM25 and TF-IDF chunk packing by 2.1 and 1.9 F1 (two-model averages, all $p < 0.001$). Since SENTPACK outperforms the pair packer in every setting of Table 1, these baselines are dominated by SENTPACK as well; we report them here rather than in the main table because they were run in a different environment.

| Model | Method | Qasper | MF-en | HotpotQA | 2Wiki | MuSiQue | NarrQA | MH | Avg. |
|---|---|---|---|---|---|---|---|---|---|
| Qwen2.5-7B | Full context | 44.86 | 51.41 | 58.07 | 45.61 | 29.36 | 28.94 | 44.35 | 43.04 |
| | LLMLingua-2 | 33.23 | 31.00 | 32.80 | 29.19 | 13.19 | 8.16 | 25.06 | 24.60 |
| | Pair packing | 40.40 | 50.08 | 54.05 | 46.25 | 26.83 | 17.05 | 42.38 | 39.11 |
| | BGE chunks | 38.90 | 47.90 | 51.61 | 45.88 | 30.72 | 18.08 | 42.74 | 38.85 |
| | SENTPACK (BGE) | 42.03 | 46.40 | 57.00 | 52.41 | 29.89 | 16.82 | 46.43 | 40.76 |
| | SENTPACK (hybrid) | 38.73 | 49.31 | 57.10 | 48.32 | 33.39 | 16.97 | 46.27 | 40.64 |
| Llama-3.1-8B | Full context | 45.28 | 54.80 | 55.79 | 44.14 | 31.26 | 27.52 | 43.73 | 43.13 |
| | LLMLingua-2 | 34.31 | 33.35 | 37.35 | 31.53 | 20.40 | 13.61 | 29.76 | 28.43 |
| | Pair packing | 38.33 | 50.09 | 49.47 | 45.11 | 24.25 | 19.46 | 39.61 | 37.79 |
| | BGE chunks | 40.65 | 48.49 | 47.33 | 48.11 | 28.50 | 20.86 | 41.31 | 38.99 |
| | SENTPACK (BGE) | 41.79 | 49.78 | 56.18 | 49.58 | 31.84 | 20.85 | 45.87 | 41.67 |
| | SENTPACK (hybrid) | 41.28 | 52.99 | 51.86 | 49.42 | 30.37 | 19.95 | 43.89 | 40.98 |
| Qwen2.5-14B | Full context | 45.46 | 53.86 | 62.00 | 58.52 | 37.21 | 28.50 | 52.58 | 47.59 |
| | LLMLingua-2 | 30.93 | 32.71 | 38.34 | 34.15 | 18.34 | 8.12 | 30.27 | 27.10 |
| | Pair packing | 39.37 | 49.88 | 56.88 | 52.25 | 31.82 | 19.92 | 46.98 | 41.69 |
| | BGE chunks | 39.75 | 49.05 | 55.42 | 53.34 | 35.28 | 21.19 | 48.02 | 42.34 |
| | SENTPACK (BGE) | 41.29 | 50.37 | 60.59 | 60.29 | 37.33 | 21.18 | 52.74 | 45.18 |
| | SENTPACK (hybrid) | 40.70 | 53.03 | 58.78 | 55.52 | 36.65 | 21.24 | 50.32 | 44.32 |

Table 6: Per-task F1 at 1,024 tokens (full context: up to 30K tokens).

| Model | Comparison | 512 | 1,024 | 2,048 |
|---|---|---|---|---|
| Qwen2.5-7B | SENTPACK − BGE chunks | +3.69 [1.69, 5.68] | +1.79 [−0.08, 3.65] | +2.62 [0.82, 4.49] |
| | SENTPACK − Pair packing | +2.86 [1.09, 4.59] | +1.53 [−0.14, 3.17] | +2.92 [1.26, 4.58] |
| | SENTPACK-BGE − BGE chunks | +2.59 [0.91, 4.29] | +1.91 [0.21, 3.64] | +2.17 [0.43, 3.96] |
| Llama-3.1-8B | SENTPACK − BGE chunks | +3.17 [1.14, 5.17] | +1.99 [0.20, 3.84] | +1.93 [0.08, 3.80] |
| | SENTPACK − Pair packing | +2.39 [0.69, 4.07] | +3.19 [1.48, 4.89] | +3.18 [1.53, 4.80] |
| | SENTPACK-BGE − BGE chunks | +2.89 [1.24, 4.56] | +2.68 [1.06, 4.38] | +2.73 [1.03, 4.44] |
| Qwen2.5-14B | SENTPACK − BGE chunks | +4.17 [2.27, 6.12] | +1.98 [0.20, 3.85] | +2.21 [0.45, 3.99] |
| | SENTPACK − Pair packing | +3.71 [2.15, 5.26] | +2.63 [1.17, 4.12] | +3.27 [1.65, 4.90] |
| | SENTPACK-BGE − BGE chunks | +2.95 [1.38, 4.53] | +2.84 [1.25, 4.49] | +2.81 [1.12, 4.59] |

Table 7: Per-model paired bootstrap differences (F1, 95% intervals). SENTPACK denotes the frozen hybrid configuration.